

TB2 mission decision

GO

Atmosphere simplification

0.06 % max diff

Fuel used

45.97 kg

Fuel remaining

194.04 kg

Mission time

6.20 hr

Path length

782.59 km

CO2 estimate

145.25 kg

BSFC scale

1.2493

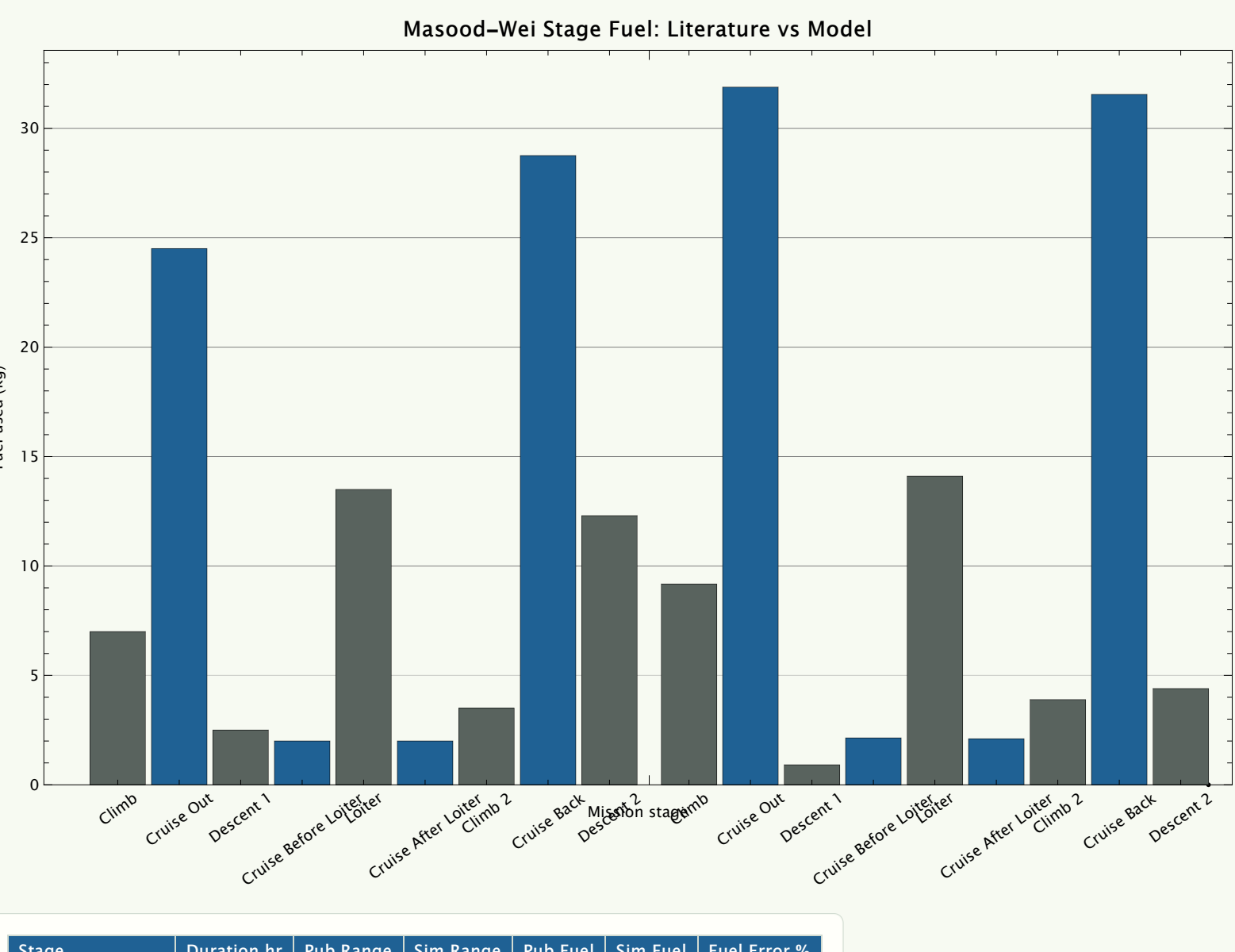
TB2: FIXED ATMOSPHERE ACCEPTED

Maximum key-output difference: 0.06%

Interpretation: For the TB2 calibrated mission, fixed atmosphere and altitude-varying atmosphere produce nearly identical mission-level outputs. For early-stage ICE/hybrid/electric comparison, keeping fixed atmosphere is justified for this mission profile.

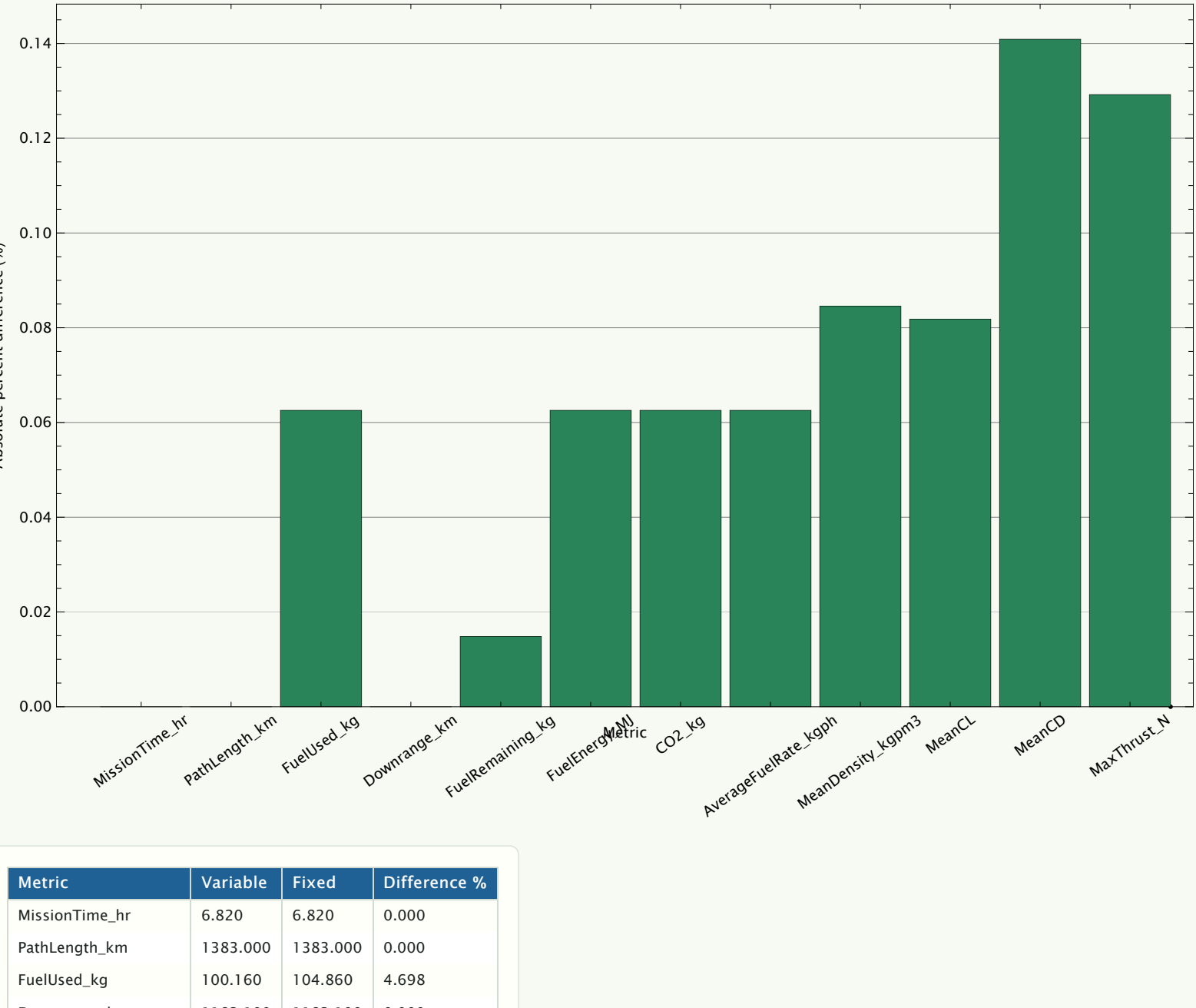
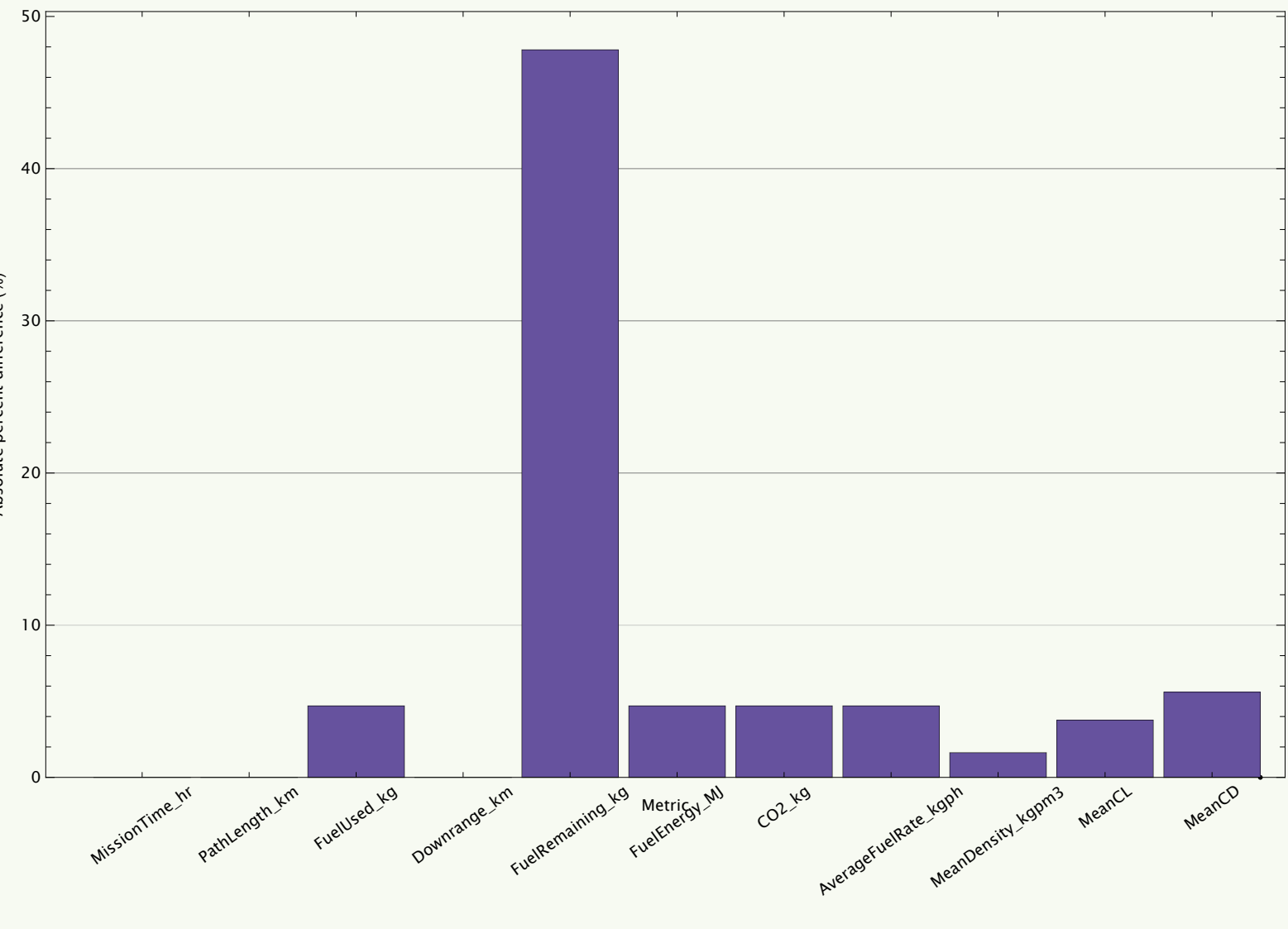
1. Paper Calibration

Metric	Published	Uncalibrated	Calibrated	Error %
Total Time (hr)	6.820	6.820	6.820	0.000
Total Range (km)	1184.000	1183.100	1183.100	-0.076
Fuel Used (kg)	101.000	80.848	100.160	-0.834
Remaining Fuel (kg)	9.000	29.152	9.842	9.357
Average Fuel Rate (kg/hr)	14.809	11.854	14.686	-0.834



Stage	Duration hr	Pub Range	Sim Range	Pub Fuel	Sim Fuel	Fuel Error %
Climb	0.410	54.000	53.546	7.000	9.175	31.064
Cruise Out	2.040	450.000	448.780	24.500	31.884	30.140
Descent 1	0.170	28.000	27.944	2.500	0.912	-63.512
Cruise Before Loiter	0.150	30.000	29.990	2.000	2.140	6.980
Loiter	1.000	0.000	0.283	13.500	14.104	4.471
Cruise After Loiter	0.150	30.000	29.780	2.000	2.100	5.020
Climb 2	0.200	30.500	30.365	3.500	3.896	11.309
Cruise Back	1.880	450.000	451.180	28.750	31.548	9.731
Descent 2	0.820	111.500	111.240	12.300	4.400	-64.227

2. Atmosphere Sensitivity



Metric	Variable	Fixed	Difference %
MissionTime_hr	6.820	6.820	0.000
PathLength_km	1383.000	1383.000	0.000
FuelUsed_kg	100.160	104.860	4.698
Downrange_km	1183.100	1183.100	0.000
FuelRemaining_kg	9.842	5.137	-47.807
FuelEnergy_MJ	4306.800	4509.100	4.698
CO2_kg	316.500	331.370	4.698
AverageFuelRate_kgph	14.686	15.376	4.698
MeanDensity_kgpm3	0.650	0.639	-1.628
MeanCL	0.572	0.593	3.762
MeanCD	0.051	0.054	5.602

Metric	Variable	Fixed	Difference %
MissionTime_hr	6.198	6.198	0.000
PathLength_km	782.590	782.590	0.000
FuelUsed_kg	45.965	45.936	-0.063
Downrange_km	290.710	290.710	0.000
FuelRemaining_kg	194.040	194.060	0.015
FuelEnergy_MJ	1976.500	1975.200	-0.063
CO2_kg	145.250	145.160	-0.063
AverageFuelRate_kgph	7.416	7.412	-0.063
MeanDensity_kgpm3	0.919	0.919	-0.085
MeanCL	1.270	1.269	-0.082
MeanCD	0.077	0.076	-0.141
MaxThrust_N	1376.300	1378.100	0.129

3. TB2 Mission Performance

